

Questions by topic: Habitat case studies: woodland, rainforest, reefs, islands, mangroves, Antarctica - Set 2 (60 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

0 1 . **1** Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

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0 1 . **2** Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

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0 1 . 3

Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

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Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

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0	2	.	1
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Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

0	2	.	2
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Explain **four** reasons why mangroves are important.

[4 marks]

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0 2 . 3 Describe the main threats to mangroves and the conservation efforts used.
[3 marks]

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0 3 Figure 1 shows a simplified Antarctic food web.

Figure 1

Trophic level	Organisms
Producers	Phytoplankton, ice algae
Primary consumers	Zooplankton, krill
Secondary consumers	Small fish, squid, baleen whales, crabeater seals, penguins
Higher consumers	Toothfish, albatross, leopard seal, killer whale

0 3 . 1 Explain why the sea around Antarctica is so productive despite the cold.
[3 marks]

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0 3 . 2

Use Figure 1 to explain why fishing for krill and for Patagonian toothfish threatens other species.

[3 marks]

0 3 . 3

Explain the importance of Antarctica in controlling global natural cycles.

[3 marks]

0 3 . 4

Explain why the area of winter sea ice around Antarctica has increased in recent years even though air and sea are warming.

[1 mark]

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0 4 . 1

Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change

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Ozone depletion

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Tourism

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Future mineral exploitation

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0 4 . 2

Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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0 5 . 1

Which of the following habitats has low biodiversity but a high proportion of endemic species? Tick one box.

[1 mark]

- A Tropical rainforest ☐
- B Oceanic islands ☐
- C Temperate broadleaf woodland ☐
- D Mangroves ☐

0 5 . 2

Compare the ecological features of tropical rainforest and Antarctica that explain their very different biodiversity.

[4 marks]

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0 6 . 1

Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

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06 . 2

Explain why growing coral reefs help to resist climate change.

[2 marks]

06 . 3

Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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END OF QUESTIONS